Semi synthetic penicillin is one in which, both fermentation and chemical approach are used to produce useful pencillins. It can be taken orally and active against gram negative bacteria. (eg) Amphicilin. Nowadays, semi synthetic pencillins makeup the bulk of the penicillin market.



The main objective of producing semi synthetic penicillin is to generate compounds with improved properties. (eg) acid stability, Resistance to enzymic degradation, broader spectrum of activity. Here side chain is removed to form (6- APA) via immobilization in a column of penicillin acylase. Penicilln G is converted to (6-APA) and phenyl acetic. Then it is chemically to acylated produce Semi Synthetic Pencillin.

New kinds of synthetic penicillin can also be produced which are readily absorbed by the intestine compared to natural penicillin. Example: Phenithicilin.

The initial strain of Penicillium chrysogenum (NRRL, 1951) was low yielding strain and so it is was treated with mutagenic agents such as X-rays, UV right and some other repeated methods to get a high yielding strain Q-176.

Production methods

Penicillin production is done by one of the following.

1. Surface culture
2. Submerged fermentation process

Inoculum Production

Inoculation methods

To inoculate fermentation medium one of the following methods can be employed.

1. Using dry spores to seed the fermentation medium.
2. Making suspension along with non toxic wetting agent like Sodium lauryl sulphate and inoculating germinated organism
3. Using pellet inocula obtained by the germination of spores

The lyophilized spores (or) spores in well sporulated frozen agar slant are suspended in water or in a dilute solution of a nontoxic wetting agent.

(1:10,000 sodium lauryl sulphonate)

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Spores are then added to a bottles containing wheat bran solution
It is incubated for 5-7 days at 24°C for heavy sporulation.

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The resulting spores are then transferred to production tank

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The micro organism in the inoculum tank is checked for contamination.

Production process

The production tanks are inoculated with a mycelial growth.

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Production medium contains following medium components.

Carbon source as Lactose, Nitrogen source as Ammonium sulphate, Acetate or Lactate (Corn steep liquor is the cheap and easy source of nitrogen)

Mineral sources as K, P (Potassium di hydrogen phosphate), Mg, S (Magnesium sulphate), Zn, Cu(Copper sulphate) (Corn steep liquor supply some of these minerals)

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